

GANs and Diffusion Models

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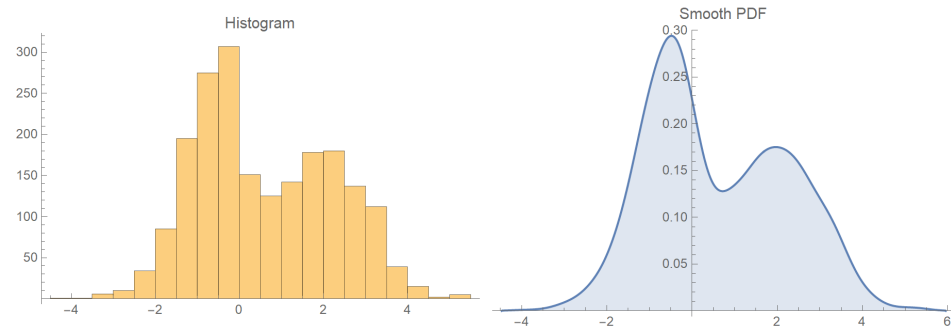
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Last time: Intro Generative Models

- Generative vs. Discriminative Models
 - **Discriminative**: Restricted to certain tasks (i.e., classification & regression)
 - **Generative**: Helpful for imperfect data (e.g., missing)
- Two key elements: **Representation & Learning**
 - **Representation**: How to model the distribution of complex data?
 - **Learning**: What is the right way to compare distributions?
- Evaluating Generative Models
 - For image: Inception Scores, Fréchet Inception Distance (FID), etc.
 - For text: coming later...

Generative Models vs Traditional Stat

- Generative model is for (high-dim) sample generation
- Traditional stat focuses on (low-dim) density estimation

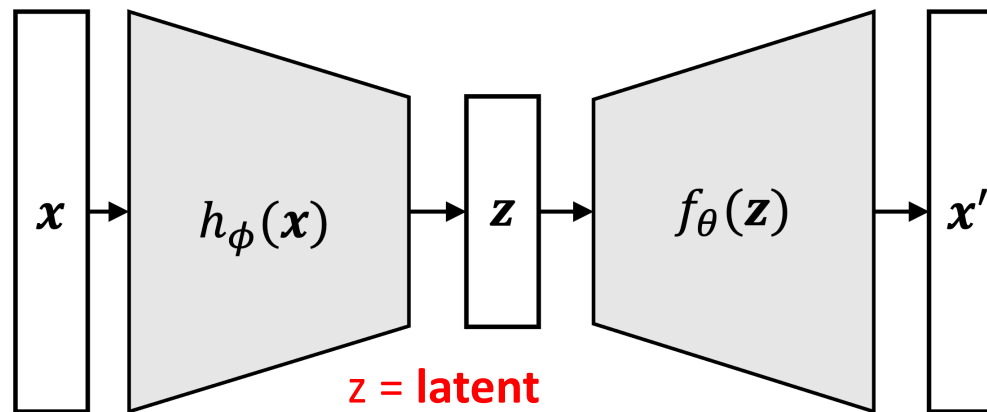


E.g., Kernel Density Estimator (KDE) can estimate density, but cannot sample...

Recap on VAE

- Distribution Representation: feed-forward neural network (FFNN)
- Structure: **Encoder** network h_ϕ + **Decoder** network f_θ

Say, an image
of $28 \times 28 = 784$
dimensions



$z = \text{latent}$
(compressed) data,
say 50 dimensions

Recap on VAE Training

- Goal: maximize (log-)likelihood:

$$\log p_{\theta}(x_1, \dots, x_N) = \sum_i \log p_{\theta}(x_i)$$

- Practical algorithm: maximize the **ELBO**

$$\log p_{\theta}(x) \geq ELBO = \int (\log p_{\theta}(x, z) - \log q_{\phi}(z|x)) q_{\phi}(z|x) dz$$

- **Yet: some disadvantages of ELBO**

1. Inconsistent estimator (introducing asymptotic bias)
2. Samples tend to have lower quality...

Rethinking the objective...

- Goal: maximize (log-)likelihood:

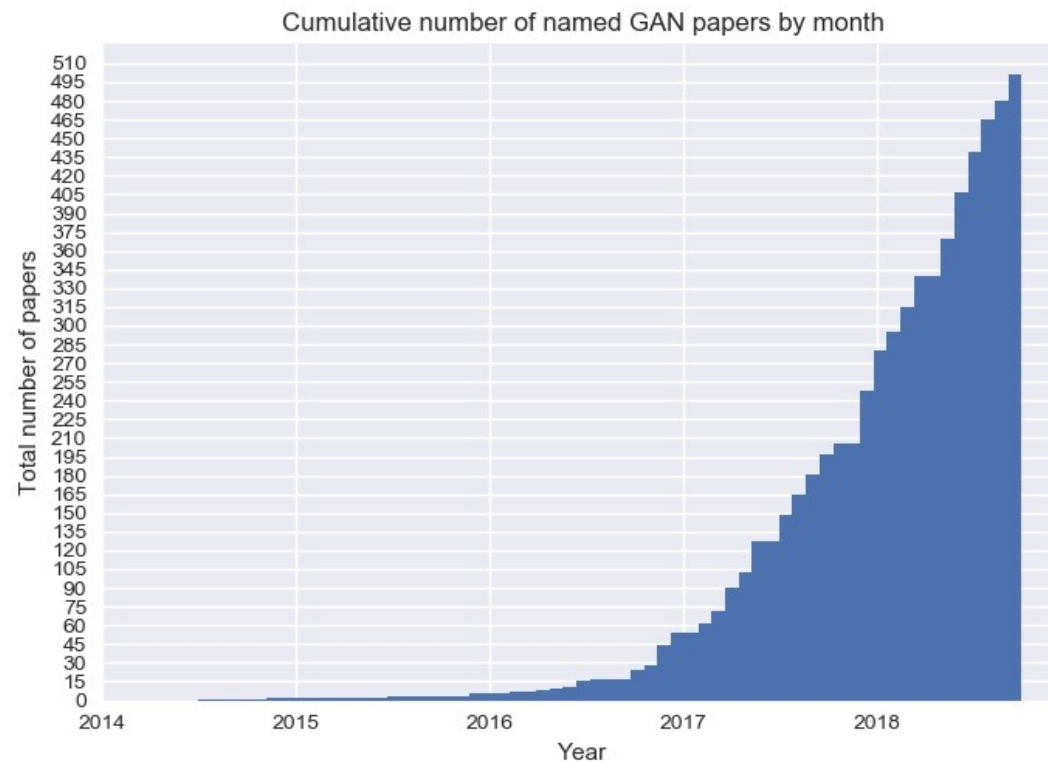
$$\max_{\theta} \mathbb{E}_{X \sim p_{data}} [\log p_{\theta}(X)]$$

- **Optimal** generative model gives the **best** sample quality and **highest** likelihood
- However, perturbations are magnified for imperfect models
 - Some examples: memorizing training data -> great samples, zero likelihood on test dataset

Agenda

1. **Generative Adversarial Networks (GANs)**
2. Diffusion models

The GAN Zoo...



Sample Video from StyleGAN3



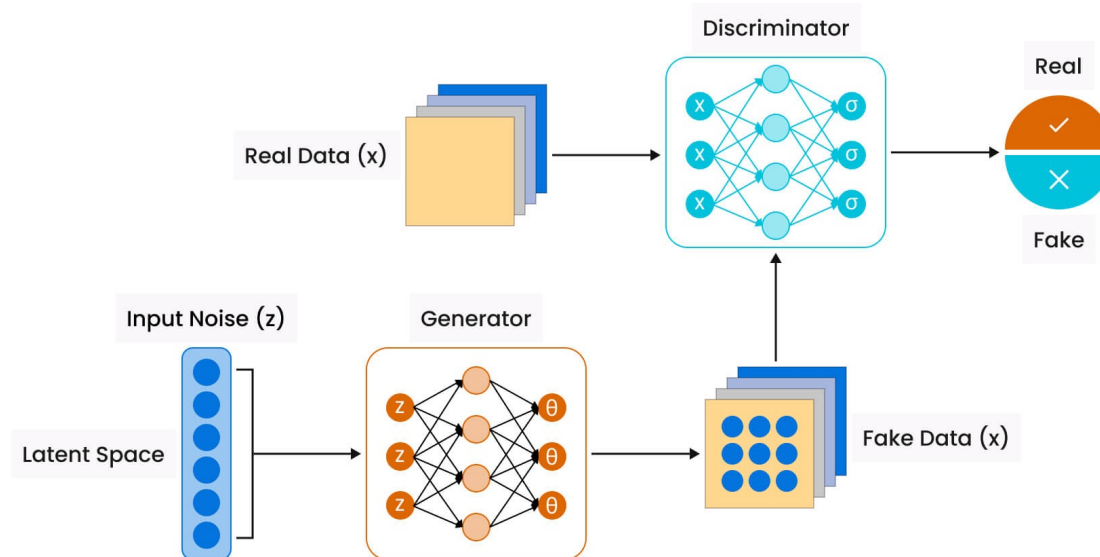
GAN: Basic Structure

- GAN is the only **likelihood-free** generative model!
- **Discriminator**: Given an input image x , output the prob that it is real
 - If x is real, $D(x) \approx 1$.
 - If x is artificial, $D(x) \approx 0$.
- **Generator**: Generate some $x = G(z)$ such that $D(G(z)) \approx 1$.
 - z is the initial latent noise (e.g., unit Gaussian)



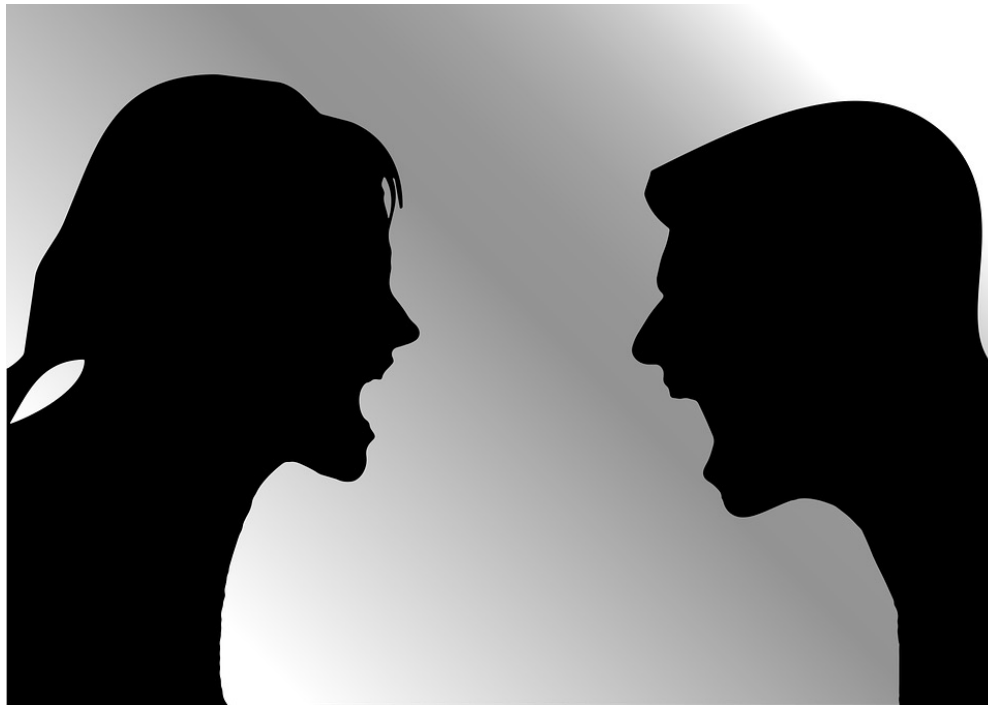
GAN: Basic Structure

Generative Adversarial Network (GAN)



Intuition of GAN: Adversarial

Generator: why
don't you follow
my suggestion??



Discriminator: why
don't you suggest
something
useful??

GAN: Training Objective

- Loss for **Discriminator**: For fixed G,

$$\max_D \left\{ \underbrace{\mathbb{E}_{x \sim p_{data}} [\log D(x)]}_{\text{Higher D values for real data}} + \underbrace{\mathbb{E}_z [\log (1 - D(G(z)))]}_{\text{Lower D values for fake data}} \right\}$$

- Loss for **Generator**: For fixed D,

$$\min_G \left\{ \mathbb{E}_z [\log (1 - D(G(z)))] \right\}$$

Can we solve GAN's optimization problem?

- Optimization problem:

$$\min_G \max_D V(G, D) = \mathbb{E}_{x \sim p_{data}} [\log D(x)] + \mathbb{E}_{y \sim p_G} [\log(1 - D(y))]$$

- For fixed G , take derivative w.r.t. D :

$$0 = \frac{\delta}{\delta D} V(G, D) = \int p_{data}(x) \frac{1}{D(x)} dx + \int p_G(x) \frac{-1}{1 - D(x)} dx \quad \text{(Rule of derivative of log)}$$

$$\implies 0 = p_{data}(x) \frac{1}{D(x)} - p_G(x) \frac{1}{1 - D(x)} \quad \text{(b/c everything is non-negative)}$$

$$\implies D^*(x) = \frac{p_{data}(x)}{p_{data}(x) + p_G(x)} \quad \text{(Rearrange the terms)}$$

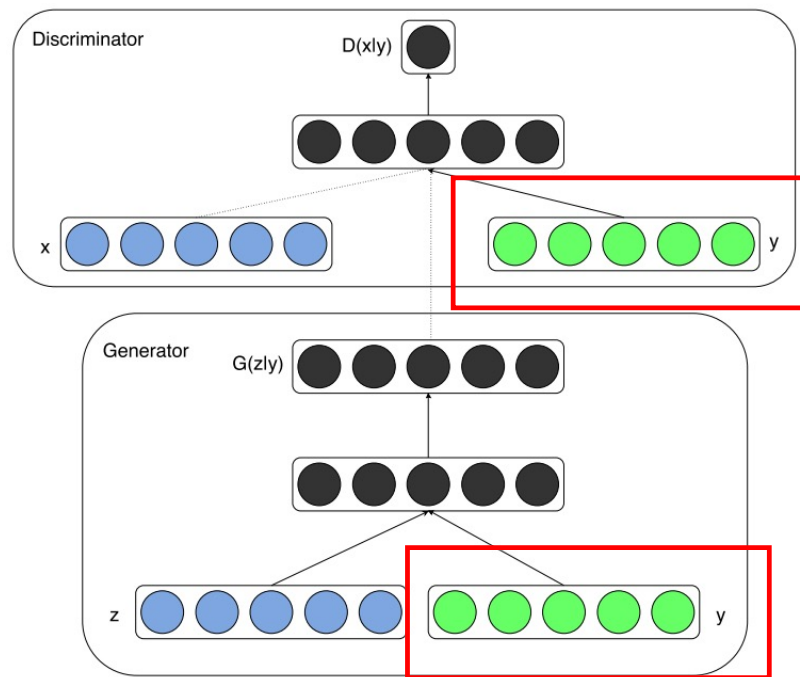
Can we solve GAN's optimization problem? (cont.)

- The optimal training loss for fixed G :

$$\begin{aligned} V(G, D^*) &= \int p_{\text{data}}(x) \log \frac{p_{\text{data}}(x)}{p_{\text{data}}(x) + p_G(x)} dx \\ &\quad + \int p_G(x) \log \frac{p_G(x)}{p_{\text{data}}(x) + p_G(x)} dx \\ &= \text{JSD}(p_{\text{data}}(x), p_G(x)) \end{aligned}$$

- JSD stands for **Jensen-Shannon Divergence**, measuring diff of dist's
- Optimal Generator: $p_G^* = p_{\text{data}}$, and $V^* = -\log 4$

Extension 1: Conditional GAN (CGAN)



Extension 2: Deep Convolutional GAN (DCGAN)

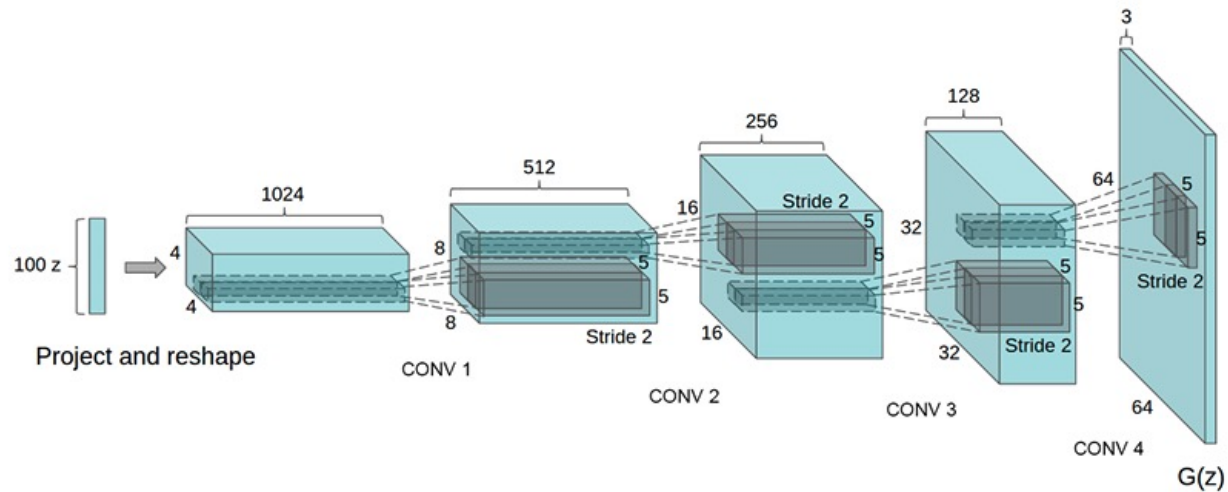


Figure 1: DCGAN generator used for LSUN scene modeling. A 100 dimensional uniform distribution Z is projected to a small spatial extent convolutional representation with many feature maps. A series of four fractionally-strided convolutions (in some recent papers, these are wrongly called deconvolutions) then convert this high level representation into a 64×64 pixel image. Notably, no fully connected or pooling layers are used.

DCGAN Simulation

https://colab.research.google.com/drive/1pChdKaxL0ZhMJA_jxpD0k2WqE1C9yCf_?usp=sharing

Some Issues with GAN

1. Vanishing gradient
 2. Unstable optimization, even non-convergence
 3. Mode collapse
- Thus, many (many) tricks:¹
 - Initialize with Gaussian rather than Uniform
 - Avoid ReLU and MaxPool as they have sparse gradients
 - Add noise to inputs and let them decay
 - Don't balance loss via statistics

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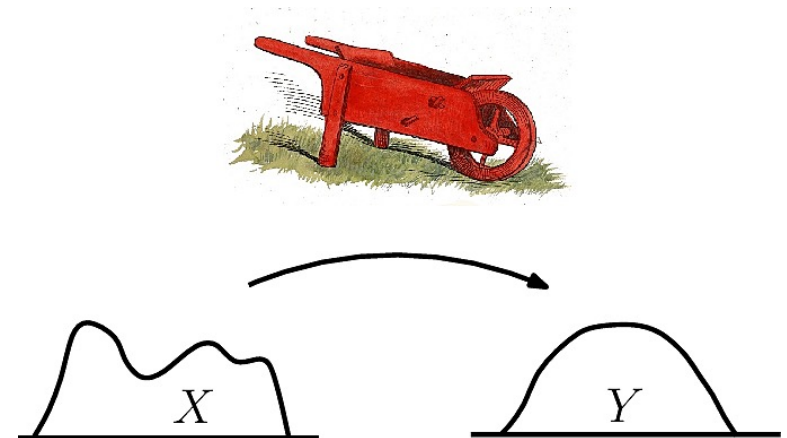
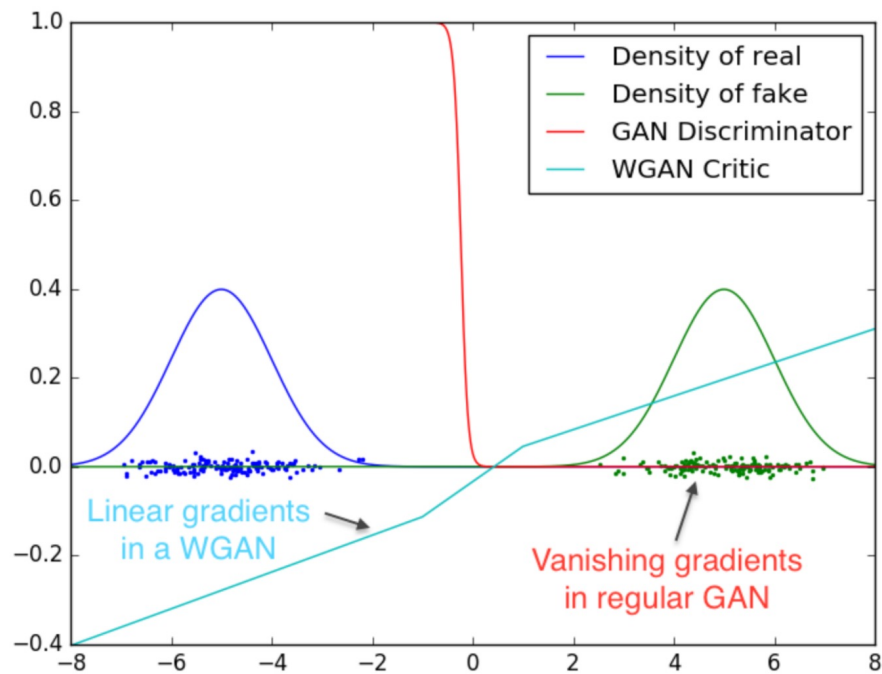
GAN: Vanishing Gradient

- Recall the loss:

$$\min_G \max_D V(G, D) = \mathbb{E}_{x \sim p_{data}} [\log D(x)] + \mathbb{E}_{y \sim p_G} [\log(1 - D(y))]$$

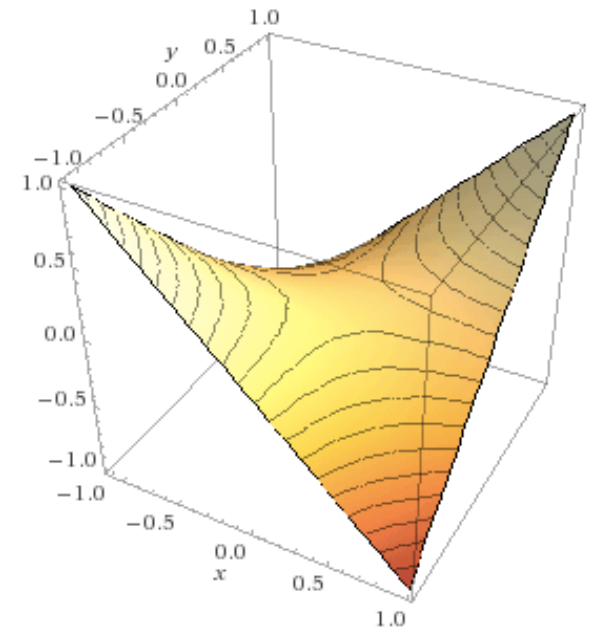
- Suppose we have a **nearly perfect discriminator**: $D(y) \approx 0$
- Then the second term (and its gradient) would be ≈ 0 , **regardless of generator!**
- This is called “vanishing gradient”

Solution: Wasserstein GAN

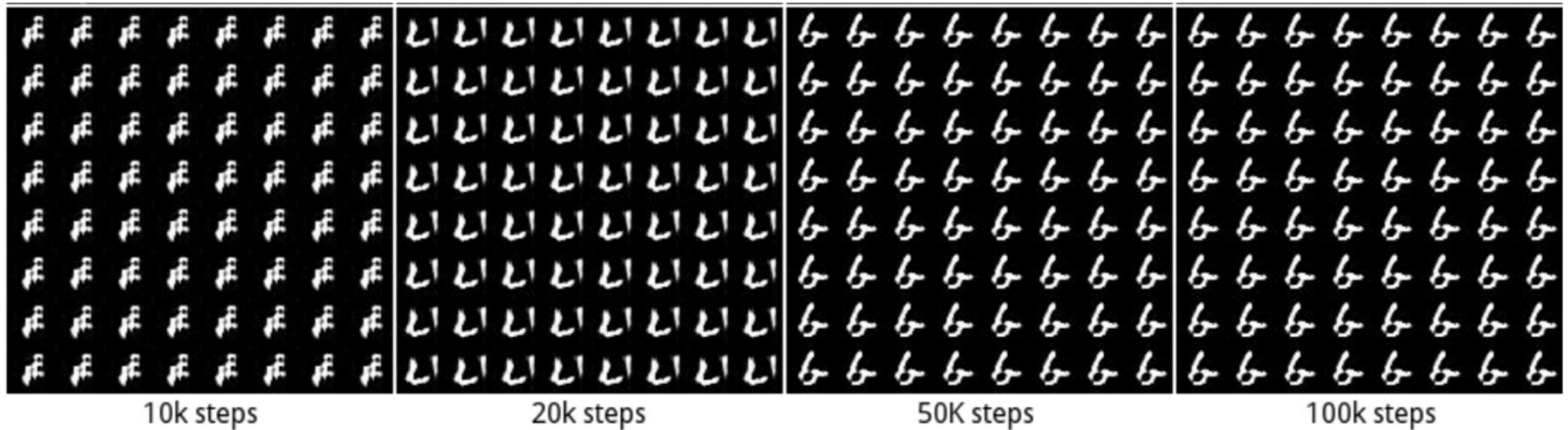


Real Issue: Non-convergence

- GAN objective: $\min_x \max_y V(x, y)$
- Consider $V(x, y) = xy$
- Equilibrium (saddle point) at $x = y = 0$
- Using gradient descent results in a **spiral**
$$\frac{\partial x_t}{\partial t} = -\frac{\partial V(x_t, y_t)}{\partial x_t}, \quad \frac{\partial y_t}{\partial t} = \frac{\partial V(x_t, y_t)}{\partial y_t}$$
$$x_t = x_0 \cos t - y_0 \sin t$$
$$y_t = x_0 \sin t + y_0 \cos t$$



Yet another convergence issue: Mode collapse

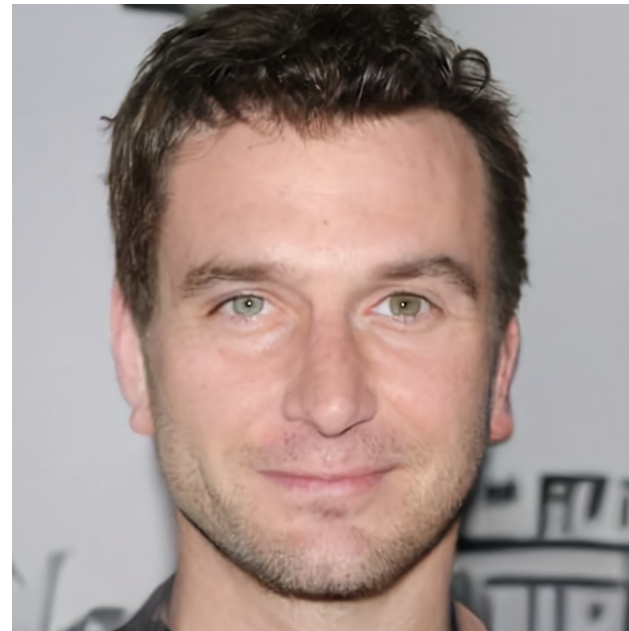


Source: Metz et al., 2017

Agenda

1. Generative Adversarial Networks (GANs)
- 2. Diffusion models**

High-Fidelity Generation for 1024x1024 Images



Diffusion Models: Motivation

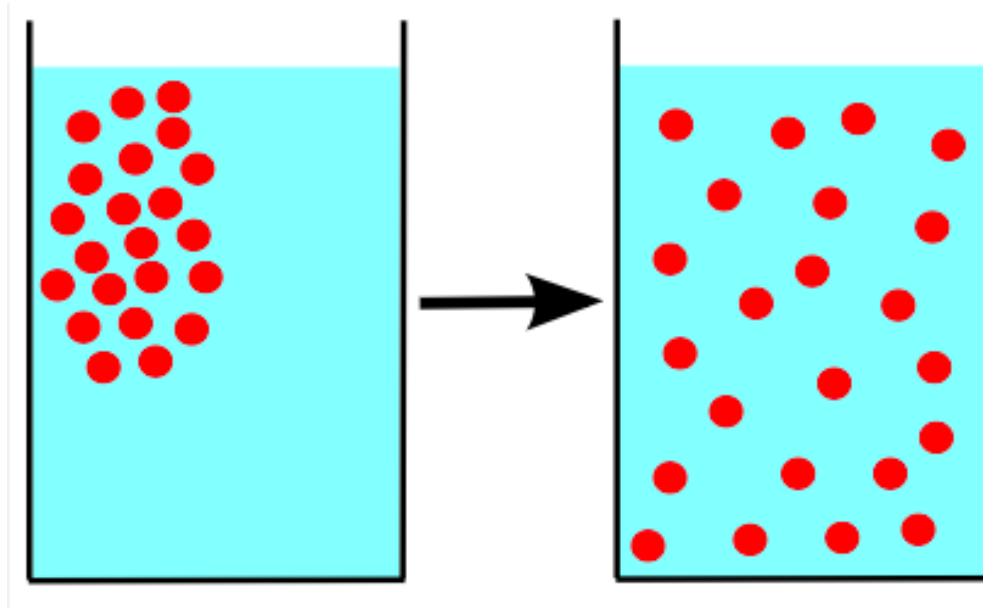
what are the typical steps of drawing a picture for a human?

The typical steps a human follows when drawing a picture usually follow a loose structure that blends planning, sketching, refining, and finishing. While the exact process varies depending on the artist's style and medium, here's a general breakdown:

Diffusion Models: Motivation (cont.)

- Takeaway: There is (roughly) a unified **structure** for creation
- Previously, VAE and GAN generate samples in “one-pass” (i.e., of the Feed-forward NN)
- Question: Can we divide the generation process into **small, manageable** steps?

Diffusion Process



Diffusion Model: Structure Overview

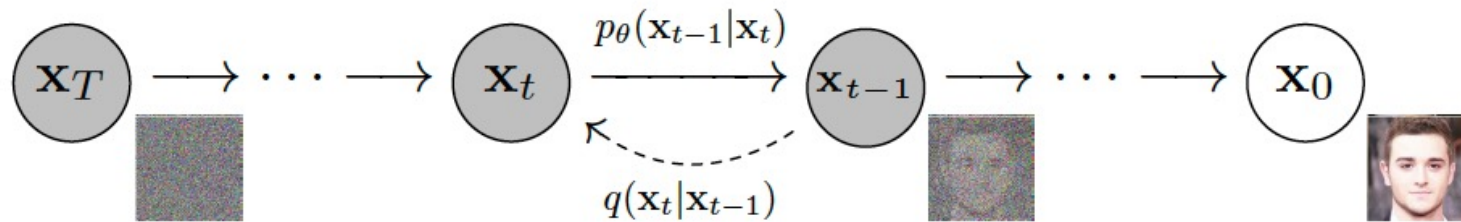


Figure 2: The directed graphical model considered in this work.

Diffusion Models related to VAEs

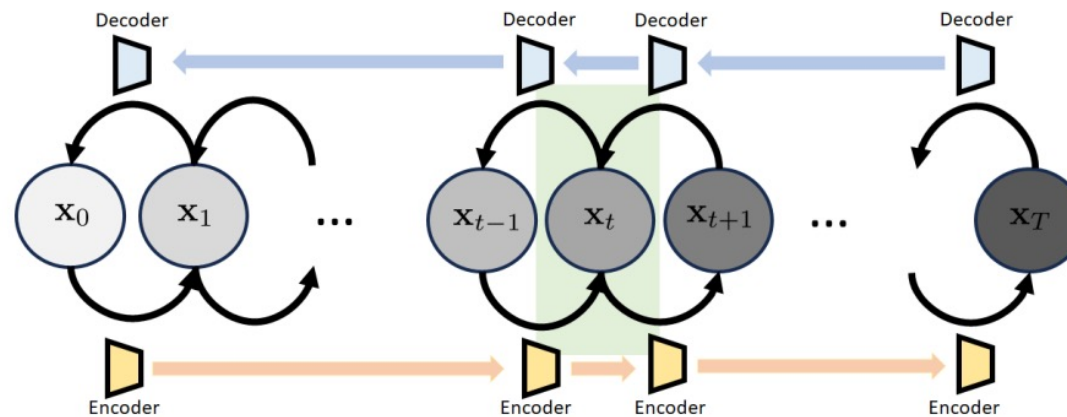


Figure 2.1: Variational diffusion model by Kingma et al [22]. In this model, the input image is $\mathbf{x}_0$ and the white noise is $\mathbf{x}_T$. The intermediate variables (or states) $\mathbf{x}_1, \dots, \mathbf{x}_{T-1}$ are latent variables. The transition from $\mathbf{x}_{t-1}$ to $\mathbf{x}_t$ is analogous to the forward step (encoder) in VAE, whereas the transition from $\mathbf{x}_t$ to $\mathbf{x}_{t-1}$ is analogous to the reverse step (decoder) in VAE. In variational diffusion models, the input dimension and the output dimension of the encoders/decoders are identical.

How to learn: Forward Process

- **Forward Process** (for noising)

$$\underset{\text{Original clean Image}}{x_0} \rightarrow x_1 \rightarrow \cdots \rightarrow \underset{\text{Veeery noisy version...}}{x_T}$$

- Divided into multiple steps (T could be 50-1000)
- During each step, a small amount of (Gaussian) noise is added:

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I)$$

“noise schedule” Gaussian noise

- Check your understanding: Does larger β_t imply faster/slower noising?

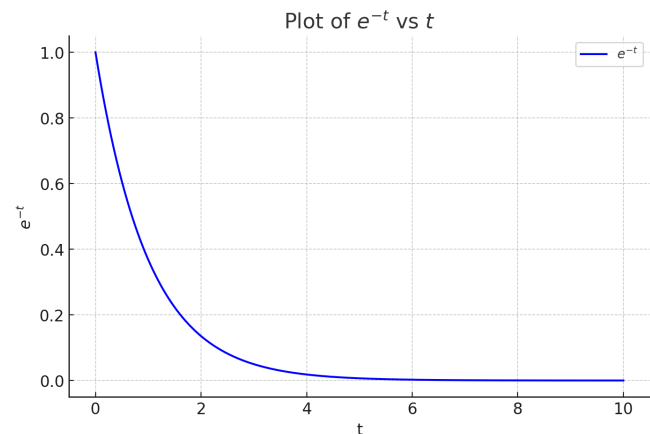
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What is the distribution of x_T ?

$$\begin{aligned}x_T &= \sqrt{1 - \beta_T}x_{T-1} + \sqrt{\beta_T}\epsilon_T \\&= \sqrt{\alpha_T}x_{T-1} + \sqrt{1 - \alpha_T}\epsilon_T \\&= \sqrt{\alpha_T\alpha_{T-1}}x_{T-2} + \sqrt{\alpha_T(1 - \alpha_{T-1})}\epsilon_{T-1} + \sqrt{1 - \alpha_T}\epsilon_T \\&= \dots \\&= \sqrt{\alpha_1 \cdots \alpha_T}x_0 + \sqrt{1 - \alpha_1 \cdots \alpha_T}\bar{\epsilon}_T \\&= \sqrt{\bar{\alpha}_T}x_0 + \sqrt{1 - \bar{\alpha}_T}\bar{\epsilon}_T\end{aligned}$$

where $\bar{\epsilon}_T$ is white Gaussian noise

Thus, we can immediately sample x_t from x_0



Sampling: Reverse Process

- **Reverse** Process (i.e., de-noising)

$$x_T \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_0$$

1. Create initial noise: $x_T \sim \mathcal{N}(0, I_d)$

2. Denoise at each step: $x_{t-1} = \frac{x_t - \beta_t \epsilon_t}{\sqrt{1 - \beta_t}}$

- Any problems?

1. We don't know the ϵ_t that produces x_t , which is **random**
2. Even though $x_t|x_{t-1}$ (conditional dist) is Gaussian, $x_{t-1}|x_t$ (**posterior dist**) is typically not

Diffusion Model: Reverse Process (cont.)

- Let's tackle the second issue first: instead of $x_{t-1}|x_t$, note that
$$q(x_{t-1}|\mathbf{x}_0, x_t) = \mathcal{N}(\tilde{\mu}_t(x_0, x_t), \tilde{\beta}_t)$$
 - $\tilde{\beta}_t$: what we have
 - $\tilde{\mu}_t(x_0, x_t)$: known function
- This is a Gaussian -> easy sampling!
- If we somehow get x_0 , we can easily sample for x_{t-1}
- How to obtain x_0 ??

Diffusion Model: Training

- We obtain x_0 by **training** for the noise $\bar{\epsilon}_t$ as a function of x_t
- For each (batch of) x_0 :
 1. Sample $\bar{\epsilon}_t \sim \mathcal{N}(0, I)$
 2. Obtain $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \bar{\epsilon}_t$
 3. Minimize $\|\epsilon_\theta(x_t) - \bar{\epsilon}_t\|^2$ (using GD/SGD/...)
 - Effectively, this maximizes the **ELBO** (Ho et al., 2020)
- During each sampling step, obtain $\hat{x}_0$ from $\epsilon_\theta(x_t)$, and use $\tilde{\mu}_t(\hat{x}_0, x_t)$ as to sample for $x_{t-1} \dots$

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Diffusion Model: Algorithms Overview

Algorithm 1 Training

- 1: **repeat**
 - 2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
 - 3: $t \sim \text{Uniform}(\{1, \dots, T\})$
 - 4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
 - 5: Take gradient descent step on
 $\nabla_{\theta} \left\| \boldsymbol{\epsilon} - \boldsymbol{\epsilon}_{\theta}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}, t) \right\|^2$
 - 6: **until** converged
-

Diffusion Model: Algorithms Overview

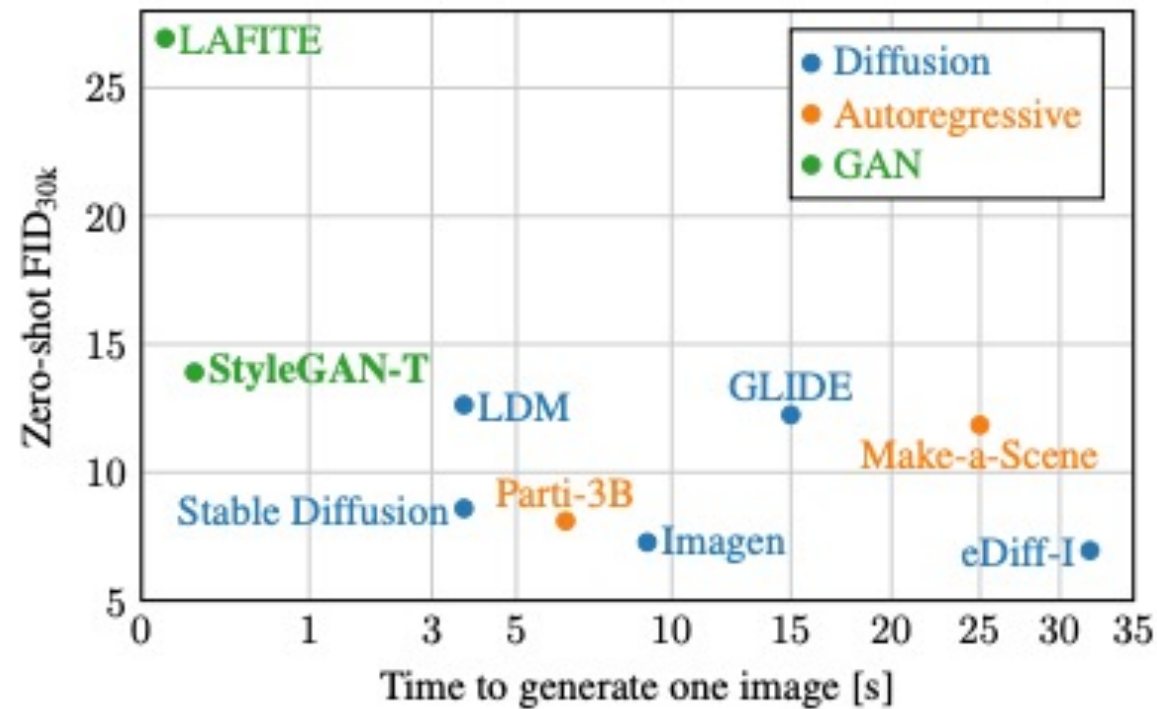
Algorithm 2 Sampling

```
1:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
2: for  $t = T, \dots, 1$  do
3:    $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = \mathbf{0}$ 
4:    $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left( \mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}$ 
5: end for
6: return  $\mathbf{x}_0$ 
```

Diffusion Model Simulation

https://colab.research.google.com/drive/1kzg-5iHQN9ZxW66uEsijLgr0w1rj2nai?usp=drive_link

Issues with Diffusion Model



Then, how to choose a model?

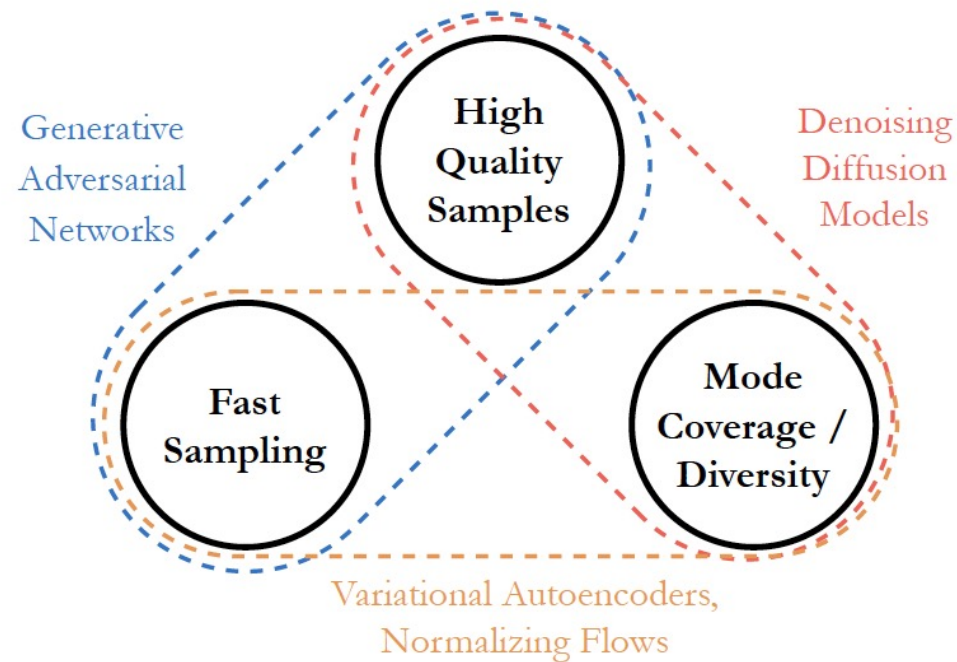
GANs

- Pros: fast sampling
- Cons: unstable training, mode collapse

Diffusion Models

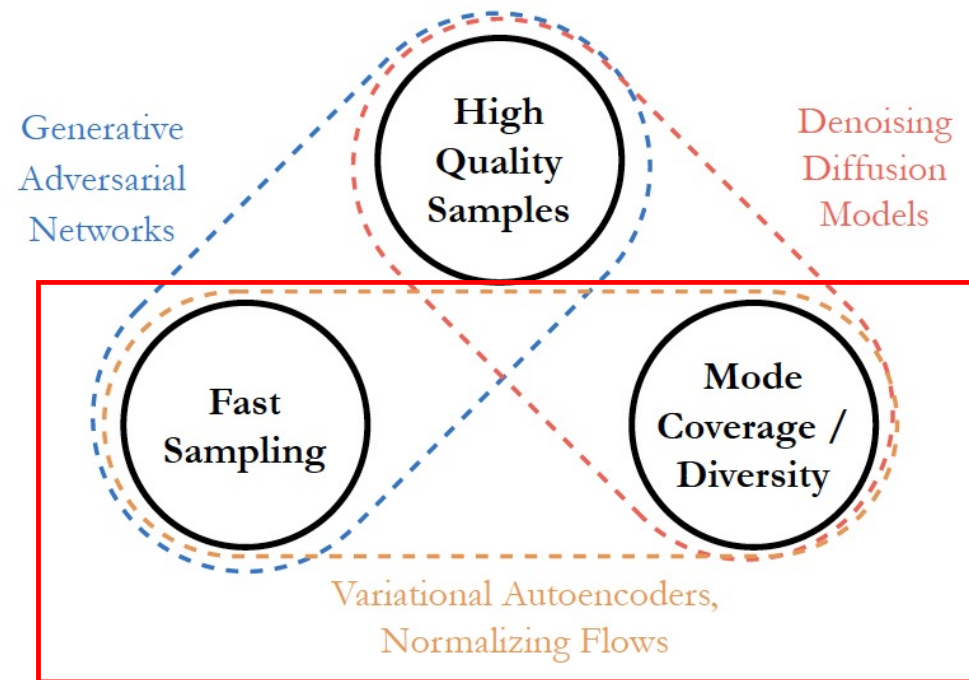
- Pros: high-quality samples, stable training
- Cons: slow!!!

Trichotomy of Existing Generative Models

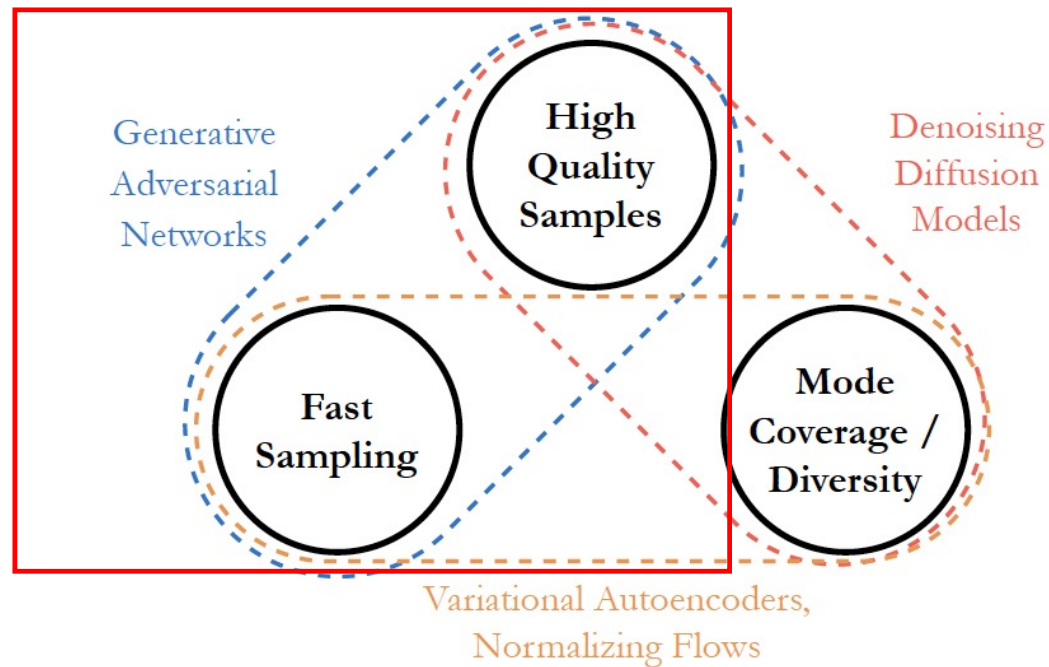


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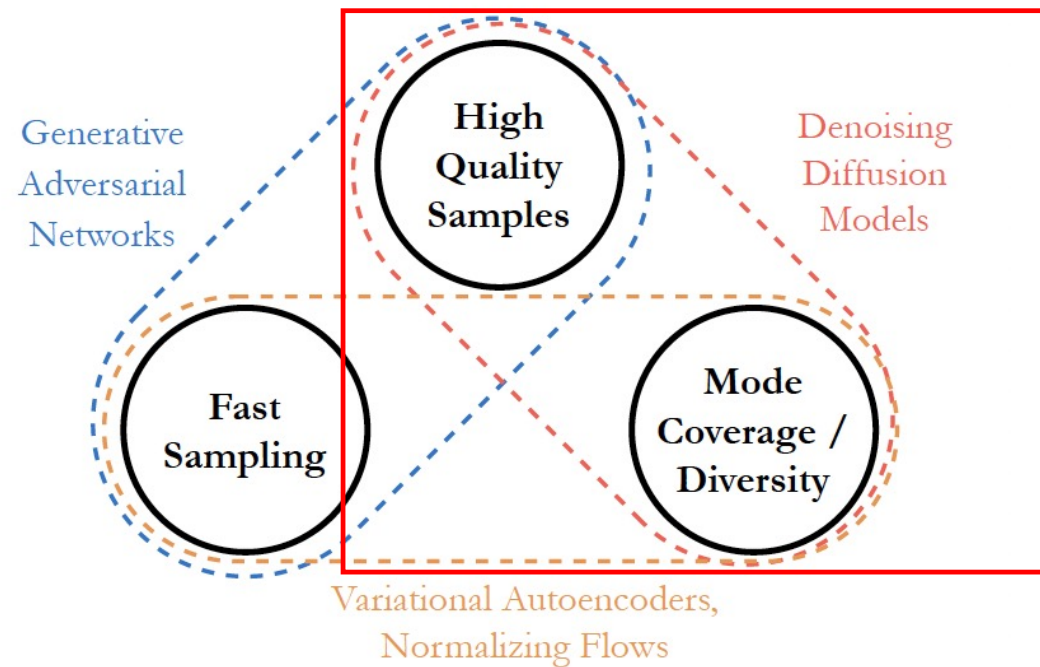
Trichotomy of Existing Generative Models



Trichotomy of Existing Generative Models



Trichotomy of Existing Generative Models



Which model would you choose?

- Entertainment and gaming?
- Medical imaging?
- Data imputation (for missing data)?
- Autonomous driving?
- Drug discovery?
- ...

Homework

Choose one of the models below, change at least 3 parameters of the model; examine any difference in the generation quality (quantitatively or qualitatively)

1. VAE
2. DCGAN
3. Diffusion Model

- Send your report to liang.1439@osu.edu

References

- Ian Goodfellow, NIPS 2016 Tutorial: Generative Adversarial Networks, <https://arxiv.org/pdf/1701.00160>
- DDPM original paper: <https://proceedings.neurips.cc/paper/2020/file/4c5bcfec8584af0d967f1ab10179ca4b-Paper.pdf>
- A blog on DDPM: <https://lilianweng.github.io/posts/2021-07-11-diffusion-models/#nice>